

ChipStain: uncertainty-aware label-free nuclear staining, towards organ-on-a-chip imaging

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Code: <https://github.com/hamidhosen42/AI4S-Open-Innovation-AI-for-Life-Science> · **Report version:** 2026-10-02 · **AI assistance:** disclosed in §10

Abstract. Nuclear read-outs on an organ-on-a-chip (cell number, proliferation, viability) start from a nuclear stain, but fixation ends the experiment, live DNA dyes add phototoxic exposure, and reporter lines are impractical for primary or iPSC-derived tissue. In-silico labeling (ISL) predicts the stain from bright-field, but a point prediction gives no warning when it is wrong. ChipStain predicts the H2B nuclear channel from one bright-field image together with a per-pixel uncertainty σ (U-Net with an ImageNet ResNet-34 encoder, β -NLL heteroscedastic head, $8\times$ test-time augmentation). We validate the uncertainty, not only the image, on a public HeLa time-lapse in a 96-well plate (held-out well, 125 images, 3 training seeds; no chip data were available to us). Against the same U-Net with TTA, ChipStain matches the correlation with the real stain (r 0.768 vs 0.764) at a small, consistent pixel-fidelity cost (MAE +10 %). Its σ ranks pixel errors better (Spearman ρ 0.47 vs 0.28; AUSE 0.177 vs 0.258; in every seed) and is approximately calibrated (nominal 95 % intervals cover 92 % of pixels). Under simulated blur and a phase-contrast input ChipStain's prediction degrades more than the plain U-Net's, but its σ rises several-fold and flags the failed images, while the U-Net's uncertainty falls. A σ gate fixed on validation data cuts the error of label-free doubling times from 9.1 % to 4.1 %. On human iPSC-derived motor neurons, fine-tuning on a single well raises Pearson r from 0.59 (zero-shot) to 0.76. For nuclei counting alone, a direct bright-field segmenter is more accurate (Cellpose F1 0.831 vs 0.708); ChipStain's value is a fluorescence-like image plus a calibrated trust signal. Open code and weights; CPU inference.

1. Problem definition and application scenario

1.1 Why a label-free nuclear channel matters on a chip

Organ-on-a-chip (OoC) systems rebuild tissue-level structure and function in microfluidic devices and are imaged repeatedly — often for days or weeks — by transmitted-light microscopy. Nearly every quantitative read-out derived from those images starts by locating nuclei: cell number, confluence, proliferation, apoptosis and pyknosis, nuclear morphology, compound dose-response. That step is usually done on a fluorescent DNA stain or a fluorescent histone reporter, each with a cost in the OoC setting:

- **Fixed stains end the experiment.** DAPI after fixation gives one time-point per chip; a time course needs one chip per time-point.
- **Live DNA dyes perturb the cells they measure.** Repeated excitation is phototoxic [1]; Hoechst 33342 is phototoxic in time-lapse imaging [11], and the live-cell dye SiR-Hoechst induces DNA-damage responses and impairs cell-cycle progression [12].
- **Reporter lines must be engineered** — routine for a cell line such as the HeLa H2B line used here, but often impractical for primary or iPSC-derived tissue, e.g. the neural cultures that are the focus of the challenge's supporting organisation.
- **Each fluorescence channel spent on nuclei** is unavailable for a functional reporter (e.g. a Ca^{2+} indicator), and every perfused reagent adds handling steps and bubble/contamination risk in small channels.

In-silico labeling (ISL) — predicting the fluorescence image from the label-free image with a neural network [2, 3] — can reduce routine staining once a paired training set exists for the system (§8), and is one of the three impact areas named by the organisers. What keeps ISL out of quantitative decisions is **trust**: a network returns an image for every input, including inputs unlike its training data, and nothing in a point prediction tells the biologist which pixels, nuclei or frames are wrong. ChipStain therefore asks a narrower and more useful question than "can we predict the stain?": **can we predict it and say — per pixel, per nucleus and per frame — how much to trust it, and does that trust signal hold up against simple alternatives?**

Target users are OoC and cell-culture laboratories and CROs running live or high-content assays, and platform developers who need a quality gate before automated analysis. **Scope and honesty statement.** All quantitative validation uses public data: 2-D HeLa monolayers in a 96-

well imaging plate (main results, 60 h time-lapse) and human iPSC-derived motor-neuron cultures in a dish (transfer test). **No organ-on-a-chip device data were available to us**; OoC is the target deployment, and §6.7 tests how the model behaves under the imaging shifts a chip introduces.

Value case (illustrative arithmetic, stated assumptions)

A neural OoC culture imaged every 30 min for 14 days at 4 fields per chip needs $14 \times 48 \times 4 = 2,688$ **nuclear-channel exposures per chip**; a label-free nuclear channel removes all of them and frees that channel for a functional reporter. Fixed DAPI would instead need one chip per time-point. Neurons are post-mitotic, so label-free nuclear counts and nuclear condensation over weeks serve as a viability/neurotoxicity end-point without repeated dye exposure. The cost of being wrong is mitigated by a frame-level σ gate (§6.10), which routes frames with unusually high mean σ to a person instead of a growth curve.

2. Related work and contribution

In-silico labeling. Christiansen et al. [2] predicted multiple fluorescent labels from transmitted-light z-stacks, outputting for every pixel a *discrete probability distribution over 256 intensity values*. Ounkomol et al. [3] (“fnet”) trained a deterministic 3-D U-Net with a mean-squared-error loss; adversarial virtual staining followed for histology [13]. **Uncertainty in bioimage restoration.** CARE [14] predicted per-pixel Laplace parameters and used the disagreement of an ensemble to flag unreliable regions; a recent diffusion-based ISL method derives hallucination maps from the variance across denoising steps [15]. **Uncertainty estimation.** Heteroscedastic regression models aleatoric noise [4]; its maximum-likelihood training is unstable and β -NLL [8] fixes this; deep ensembles [16] and test-time augmentation [17] add model-related uncertainty; sparsification curves and AUSE [18] score how well an uncertainty ranks errors.

Contribution. ChipStain introduces no new uncertainty mechanism. Its contributions are: (i) a β -NLL-stabilised heteroscedastic ISL model with test-time augmentation for bright-field $\rightarrow$ nuclei, released as an open, CPU-capable tool; and (ii) a validation of its uncertainty that the ISL literature rarely reports — against an equally strong TTA baseline, against uncertainty-free proxies (predicted intensity, edge strength, cell density), at pixel, nucleus and frame level, under imaging shift, on a biological read-out (doubling time) with a gate fixed on validation data, and on neural cultures. We report where σ helps and, equally, where it does not.

3. Data

3.1 Main dataset: source, licence, usage note

Automatic labelling of HeLa "Kyoto" cells using Deep Learning tools by Romain Guet (EPFL BioImaging & Optics Platform), Zenodo 10.5281/zenodo.6140064, CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>) [5], derived from the time-lapse *HeLa "Kyoto" cells under the scope* (10.5281/zenodo.6139958, CC BY 4.0) [6]. HeLa "Kyoto" cells stably expressing EGFP- α -tubulin and mCherry-H2B were grown as a 2-D monolayer in a CellCarrier Ultra 96-well plate (PerkinElmer) and imaged on a PerkinElmer Operetta (20 $\times$ /NA 0.8) every 15 min for 60 h. The archive provides, for 5 time-points (frames 1, 10, 50, 100, 150 $\approx$ 0–37 h) of every field, a 540 $\times$ 540 px 16-bit bright-field image, two digital-phase-contrast renderings, a two-channel fluorescence image and StarDist nuclei labels computed automatically on the H2B channel [10]. Input is bright-field; target is mCherry-H2B (channel 1: correlation with the binarised nuclei labels 0.787 vs 0.416 for tubulin over the 200 training images; `scripts/check_channels.py`). **Usage note.** The provider asks that the fluorescence not be reused *with the derived labels* for training and notes that the automatic labels are imperfect. We comply: we train only bright-field $\rightarrow$ fluorescence, never on the labels, and use the StarDist labels solely as reference objects for evaluation. Segmentation F1 and count errors therefore measure agreement with StarDist run on the real stain, not with manual ground truth.

3.2 Splits without leakage

Split	Wells	Fields	Images	Use
Train	R05-C05, R05-C07	0–19 ($\times 5$ time-points)	200	optimisation
Validation	R05-C05, R05-C07	20–24 ($\times 5$ time-points)	50	checkpoint selection; gate and recalibration thresholds
Test	R05-C03 (held-out well, same plate and imaging session)	all 25 ($\times 5$ time-points)	125	all reported numbers

All time-points of a field stay on the same side of every split. All three wells come from one plate and one imaging session (file prefix 20210904_TL2), so the hold-out measures well-to-well generalisation, not generalisation across experiments, days or microscopes. Because the 125 test images are 25 fields $\times$ 5 time-points of one time-lapse, the field ($n = 25$) — not the image — is the unit of statistical testing (§6.2).

3.3 Additional evaluation data

60 h time-lapse. The full 240-frame recording of test field R05-C03-F0 (10.5281/zenodo.6139958, CC BY 4.0) is used only for evaluation (§6.10); five of its frames are also test images. **Neural cultures.** Condition A of the in-silico-labeling data [2] (human iPSC-derived motor neurons; bright-field z-stacks and widefield DAPI; 22 training and 3 test wells; CC BY 4.0) is used for the transfer test (§6.11). **Direct-segmentation competitor.** The Cellpose models published by the dataset author (10.5281/zenodo.6140111, CC BY 4.0), trained to segment nuclei directly from bright-field (§6.8).

3.4 Pre-processing, compliance and ethics

Bright-field images are normalised per image with their 1st/99th percentiles ($\approx [-2, 2]$), making the model insensitive to camera offset and gain; the target is scaled with fixed constants from the training H2B distribution, $(I - 600)/(20\,000 - 600)$, clipped to $[0, 1]$, and all image metrics are computed in this space. Training uses random 256×256 crops with 90° rotations and flips; evaluation uses full 540×540 images. Figures show percentile-normalised, colour-mapped renderings; every image is listed with its source and modifications in Appendix E. All data are public and CC BY 4.0. HeLa is an established human-derived cell line (taken from Henrietta Lacks in 1951 without her consent); the images contain no genetic, clinical or personal information. No clinical, personal or restricted data are used.

4. Method

4.1 Backbone

A U-Net [7] built with segmentation_models_pytorch [9]; ResNet-34 encoder [19] initialised from ImageNet [20]; five-stage decoder with skip connections; one input channel; ≈ 24.4 M parameters.

4.2 Heteroscedastic output and β -NLL

The last layer outputs two maps: the predicted mean $\mu(x)$ and log-variance $s(x) = \log \sigma^2(x)$ of a per-pixel Gaussian over the target y . The Gaussian negative log-likelihood

$$L_{NLL} = \frac{1}{2} \sum_p [s_p + (y_p - \mu_p)^2 \cdot e^{-s_p}]$$

is known to be unstable: pixels that are hard early in training receive a large σ , the gradient on their mean is down-weighted by e^{-s} , and they are under-fitted [8]. β -NLL [8] multiplies each pixel's loss by the *detached* factor $(\sigma^2)^\beta$; we use $\beta = 0.5$ ($\beta = 0$ recovers the NLL; at $\beta = 1$ the gradient with respect to μ equals the MSE gradient while σ^2 is still fitted). Because the loss is Gaussian, the mean head is trained with a σ -weighted squared error rather than the L1 loss of our baseline — a confound we test explicitly (§6.3). s is clamped to $[-10, 4]$. σ is the model's learned (aleatoric) uncertainty.

4.3 Test-time augmentation, ensembles and the reported σ

At inference the network is applied to the eight dihedral transforms of the input (4 rotations $\times$ optional flip); outputs are mapped back and $\bar{\mu}$ is their mean. The reported uncertainty is $\sigma_{\text{tot}} = \sqrt{(\text{mean of the eight } \sigma^2 + \text{variance of the eight } \mu)}$. We call the second term **TTA disagreement**: it measures how much the prediction changes under symmetries the model should respect, a partial proxy for model uncertainty, not a Bayesian epistemic estimate [17]. For the plain U-Net baseline, the TTA disagreement alone is its uncertainty. We also evaluate **3-seed deep ensembles** [16] built from the cached predictions of the three training seeds ($\sigma^2 = \text{variance of member means} + \text{mean member } \sigma^2$).

4.4 From pixels to decisions

Per nucleus: nuclei are segmented on the prediction (Gaussian blur, Otsu threshold, distance-transform watershed); each nucleus receives the mean σ inside it. **Per frame:** the mean σ of the image, divided by the median over frames of the same time-point (cell density grows with time, and σ with density); a frame is flagged when this ratio exceeds its 95th percentile **on the validation split** — a 5 % false-alarm budget fixed without test labels.

4.5 Evaluation protocol

Fidelity: MAE, PSNR, SSIM, Pearson r against the real H2B. **Downstream:** object-level F1 at IoU ≥ 0.5 against the StarDist reference nuclei; the same pipeline run on the real H2B gives the reference level a perfect prediction would reach. **Uncertainty ranking:** Spearman $\rho(\sigma, |\text{error}|)$ on a fixed 200 000-pixel sample per image; AUSE [18] (area between the sparsification curve and the oracle curve, normalised; lower is better); and the MAE drop after removing the 20 % highest- σ pixels — each also computed for two uncertainty-free proxies taken from the prediction itself: predicted intensity μ and its edge strength $|\nabla\mu|$. **Calibration:** empirical coverage of $\mu \pm \sigma$. **Statistics:** 3 training seeds per configuration. Inference rests on 95 % confidence intervals from a hierarchical bootstrap that resamples seeds and fields, and on per-seed field-level Wilcoxon tests; a difference is called robust only if its CI excludes zero, no seed goes the other way and at least two seeds are significant on their own. A pooled field-level Wilcoxon (Holm-corrected) is reported for the uncertainty metrics only.

5. Implementation

Item	Value
Software	Python 3.12.11, PyTorch 2.14.0, segmentation_models_pytorch 0.5.0, scikit-image 0.26.0, SciPy 1.18.1, NumPy 2.5.2 (exact pins: requirements-lock.txt)
Optimiser	AdamW (weight decay 1e-4), one-cycle LR (peak 1e-3 from scratch, 5e-4 with the pretrained encoder; 10 % warm-up), gradient clipping 1.0
Training	40 epochs $\times$ 2 random-crop passes over the 200 training images; batches of 8×256^2 crops; checkpoint = best validation Pearson + SSIM
Seeds	0, 1, 2 for every configuration (main and ablation arms)
Hardware / time	Apple M5 laptop (Metal/MPS): 9–28 min per 40-epoch run (measured from the run logs)
Inference	runs on a CPU (CUDA/MPS used if present): ≤ 0.5 s per 540×540 image single pass, ≤ 2.4 s with 8 $\times$ TTA (measured on CPU)
External downloads	datasets from Zenodo / Google Cloud (public); ResNet-34 ImageNet weights from the Hugging Face Hub (training only); released checkpoint from the GitHub release. No paid service, API or proprietary hardware.
Entry points	scripts/train.py, scripts/evaluate.py, scripts/inference.py, demo/app.py; analyses in scripts/ (§9)

6. Experiments and results

6.1 Main results over three training seeds

Model	Pearson r ↑	SSIM ↑	MAE ↓	PSNR ↑	seg-F1 ↑	$\rho(\sigma, \text{err})$ ↑	AUSE ↓	MAE drop, top-20 % σ removed ↑
U-Net baseline (scratch, L1, lr 1e-3)	0.749 ± 0.027	0.826 ± 0.003	0.0299 ± 0.0006	24.15 ± 0.24	0.648 ± 0.089	—	—	—
+ ImageNet encoder (L1, lr 5e-4)	0.755 ± 0.005	0.823 ± 0.005	0.0296 ± 0.0003	24.25 ± 0.07	0.669 ± 0.021	—	—	—
U-Net baseline + TTA (σ = view s.d.)	0.764 ± 0.024	0.831 ± 0.003	0.0294 ± 0.0006	24.36 ± 0.25	0.665 ± 0.083	0.283 ± 0.071	0.258 ± 0.025	40.5 ± 3.2 %
+ ImageNet encoder + TTA (σ = view s.d.)	0.768 ± 0.006	0.827 ± 0.005	0.0291 ± 0.0003	24.47 ± 0.10	0.690 ± 0.019	0.317 ± 0.037	0.248 ± 0.011	42.9 ± 1.8 %
ChipStain (β -NLL, β = 0.5)	0.757 ± 0.015	0.810 ± 0.018	0.0324 ± 0.0013	23.92 ± 0.36	0.697 ± 0.012	0.430 ± 0.006	0.303 ± 0.072	45.6 ± 1.6 %
ChipStain + TTA (full)	0.768 ± 0.012	0.814 ± 0.022	0.0323 ± 0.0007	24.11 ± 0.32	0.708 ± 0.011	0.469 ± 0.027	0.177 ± 0.021	49.7 ± 4.2 %
Real fluorescence, same segmentation pipeline (reference level, not a bound)	—	—	—	—	0.765	—	—	—

Mean $\pm$ s.d. over 3 training seeds; each seed is the mean over the 125 test images of well R05-C03. The released checkpoint is seed 0 (ChipStain + TTA: Pearson r 0.779, SSIM 0.808, seg-F1 0.718, $\rho(\sigma, \text{err})$ 0.441). The real-fluorescence row runs the same watershed on the real H2B image; it is a reference level, not a strict bound.

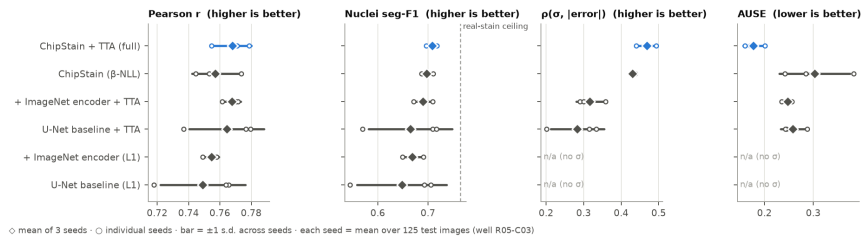


Figure 1. All configurations over three training seeds: open circles are seeds, diamonds their mean, bars ± 1 s.d. ChipStain + TTA (blue) is the full method.

6.2 Like-for-like comparison and statistics

The fair comparison for the full method is a U-Net that also uses 8× TTA (its uncertainty is then the TTA disagreement). Differences are ChipStain + TTA minus U-Net + TTA, with a 95 % CI from a bootstrap that resamples both seeds and fields, and a field-level Wilcoxon test (Holm-corrected).

- **Correlation with the real stain:** Pearson r +0.004 (95 % CI -0.021 to +0.030; significantly better in 1/3 seeds, worse in 0/3) — comparable; no robust difference.
- **Pixel fidelity:** MAE +0.0029 (95 % CI +0.0009 to +0.0054; significantly better in 0/3 seeds, worse in 2/3); SSIM -0.017 (95 % CI -0.036 to +0.006; significantly better in 0/3 seeds, worse in 2/3). The MAE cost is small but consistent (its CI excludes zero); SSIM is lower in most seeds. §6.3 tests whether it comes from the loss or the head.
- **Downstream nuclei F1:** +0.043 (95 % CI -0.017 to +0.135; significantly better in 1/3 seeds, worse in 0/3): higher on average but not robust across seeds — the U-Net's seed-to-seed spread (s.d. 0.083 vs 0.011 for ChipStain) comes from one unstable U-Net run ($n = 3$ seeds, descriptive only).
- **Uncertainty ranking:** $p(\sigma, |\text{error}|)$ +0.186 (95 % CI +0.120 to +0.264; significantly better in 3/3 seeds, worse in 0/3; better in 25/25 fields; pooled Holm $p < 0.001$); AUSE -0.081 (95 % CI -0.114 to -0.049; significantly better in 3/3 seeds, worse in 0/3; better in 25/25 fields; pooled Holm $p < 0.001$). This is the clearest and most consistent result of the study.

Matched control. The U-Net above differs from ChipStain in encoder initialisation and learning rate as well as in the head. Against an ImageNet-encoder L1 U-Net with the same TTA and learning rate — differing only in the probabilistic head and loss — ChipStain's uncertainty is still better: p +0.152 (95 % CI +0.106 to +0.201; significantly better in 3/3 seeds, worse in 0/3; better in 25/25 fields; pooled Holm $p < 0.001$); AUSE -0.070 (95 % CI -0.095 to -0.045; significantly better in 3/3 seeds, worse in 0/3; better in 25/25 fields; pooled Holm $p < 0.001$); with Pearson r +0.000 (95 % CI -0.019 to +0.013; significantly better in 2/3 seeds, worse in 0/3) and MAE +0.0032 (95 % CI +0.0012 to +0.0056; significantly better in 0/3 seeds, worse in 2/3).

Without TTA, ChipStain's learned σ alone still ranks errors better than the U-Net's TTA disagreement in p , but **not** in AUSE (0.303 vs 0.258; Holm $p = 1$): the learned head and the TTA term are complementary, and the full method needs both.

A vs B	metric	A	B	A - B (95 % CI, seeds + fields)	seeds sig. better / worse	p (pooled, Holm; σ metrics)
ChipStain + TTA vs U-Net + TTA	Pearson r	0.7681	0.7644	+0.0037 (-0.0211, +0.0304)	1 / 0 of 3	—
ChipStain + TTA vs U-Net + TTA	SSIM	0.8137	0.8310	-0.0173 (-0.0361, +0.0061)	0 / 2 of 3	—
ChipStain + TTA vs U-Net + TTA	MAE	0.0323	0.0294	+0.0029 (+0.0009, +0.0054)	0 / 2 of 3	—
ChipStain + TTA vs U-Net + TTA	psnr	24.1148	24.3614	-0.2466 (-0.7972, +0.1756)	0 / 0 of 3	—
ChipStain + TTA vs U-Net + TTA	nuclei F1	0.7082	0.6651	+0.0431 (-0.0171, +0.1351)	1 / 0 of 3	—
ChipStain + TTA vs U-Net + TTA	$\rho(\sigma, \text{error})$	0.4689	0.2832	+0.1857 (+0.1201, +0.2641)	3 / 0 of 3	2.4e-06
ChipStain + TTA vs U-Net + TTA	AUSE	0.1775	0.2582	-0.0807 (-0.1136, -0.0493)	3 / 0 of 3	2.4e-06
ChipStain + TTA vs ImageNet-L1 U-Net + TTA	Pearson r	0.7681	0.7678	+0.0003 (-0.0194, +0.0133)	2 / 0 of 3	—
ChipStain + TTA vs ImageNet-L1 U-Net + TTA	SSIM	0.8137	0.8271	-0.0134 (-0.0323, +0.0103)	0 / 2 of 3	—
ChipStain + TTA vs ImageNet-L1 U-Net + TTA	MAE	0.0323	0.0291	+0.0032 (+0.0012, +0.0056)	0 / 2 of 3	—
ChipStain + TTA vs ImageNet-L1 U-Net + TTA	psnr	24.1148	24.4676	-0.3528 (-0.9085, -0.0132)	0 / 0 of 3	—
ChipStain + TTA vs ImageNet-L1 U-Net + TTA	nuclei F1	0.7082	0.6898	+0.0184 (-0.0092, +0.0419)	2 / 0 of 3	—
ChipStain + TTA vs ImageNet-L1 U-Net + TTA	$\rho(\sigma, \text{error})$	0.4689	0.3165	+0.1523 (+0.1057, +0.2010)	3 / 0 of 3	2.4e-06
ChipStain + TTA vs ImageNet-L1 U-Net + TTA	AUSE	0.1775	0.2477	-0.0702 (-0.0947, -0.0453)	3 / 0 of 3	2.4e-06
ImageNet-L1 U-Net + TTA vs U-Net + TTA	$\rho(\sigma, \text{error})$	0.3165	0.2832	+0.0334 (-0.0389, +0.1162)	2 / 0 of 3	0.24
ImageNet-L1 U-Net + TTA vs U-Net + TTA	AUSE	0.2477	0.2582	-0.0105 (-0.0402, +0.0153)	2 / 0 of 3	0.46
ChipStain vs ImageNet-L1 U-Net + TTA	$\rho(\sigma, \text{error})$	0.4302	0.3165	+0.1137 (+0.0725, +0.1566)	3 / 0 of 3	4.1e-06
ChipStain vs ImageNet-L1 U-Net + TTA	AUSE	0.3029	0.2477	+0.0552 (-0.0213, +0.1935)	0 / 0 of 3	1
ChipStain vs U-Net + TTA	$\rho(\sigma, \text{error})$	0.4302	0.2832	+0.1470 (+0.0900, +0.2271)	3 / 0 of 3	2.4e-06
ChipStain vs U-Net + TTA	AUSE	0.3029	0.2582	+0.0447 (-0.0344, +0.1813)	0 / 0 of 3	1

Table 2. Paired comparisons, all with TTA on both sides (except single-pass ChipStain, marked). Inference rests on the hierarchical seed+field bootstrap CI and on per-seed tests; the pooled field-level p-value averages over seeds, so it is shown for the uncertainty metrics only. For MAE and AUSE lower is better.

6.3 Ablation: what each design choice does

The de-confounded ablation arms are still training; this section is completed when they finish.

6.4 Qualitative results and failure cases

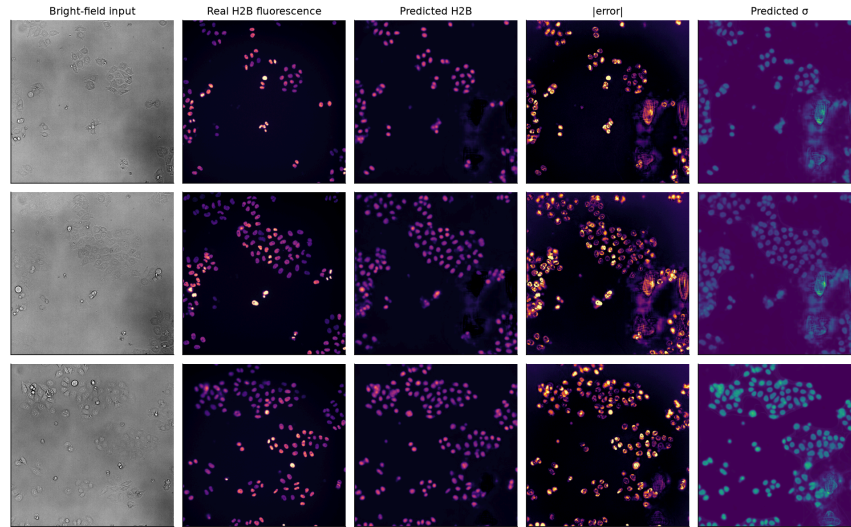
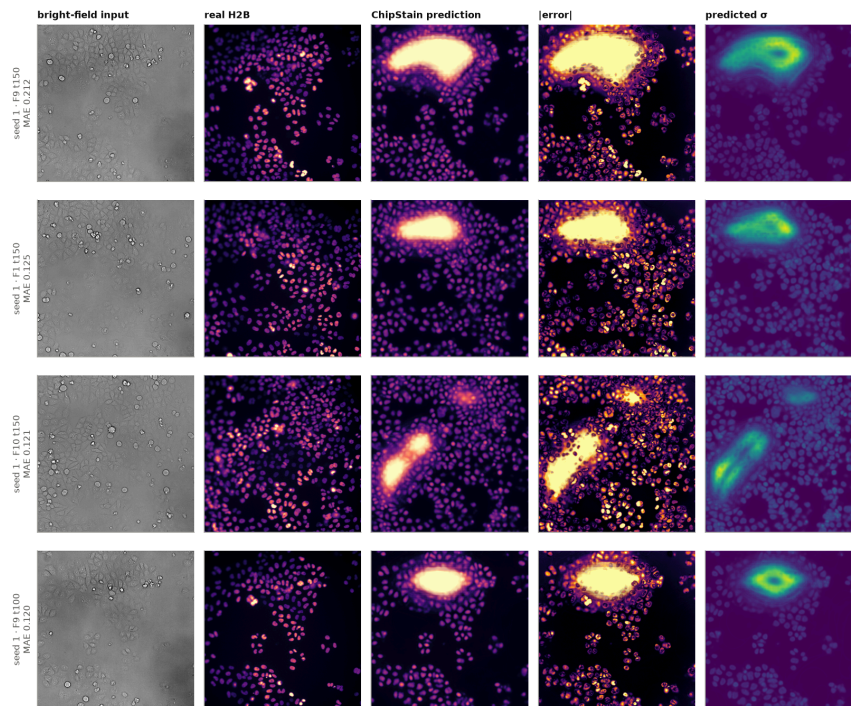


Figure 3. Test images (seed-0 checkpoint, TTA). Columns: bright-field input, real H2B, prediction, absolute error, predicted σ . σ is raised over every nucleus (heteroscedastic σ scales with intensity) and is near zero on background. The strongest σ and error hotspots are the elliptical, grid-textured patches on the right of rows 1–2 (the same field at two time-points, same position) and at the bottom right of row 3: a fixed optical/plate artefact that the model renders as dark textured regions. We did not annotate mitoses, debris or focus; nucleus-level σ is quantified in §6.9. Data: R. Guet, EPFL BIOP, Zenodo 10.5281/zenodo.6140064, CC BY 4.0.



Data: R. Guet, EPFL BIOP, Zenodo 10.5281/zenodo.6140064, CC BY 4.0

Figure 4. The worst test images by MAE across all ChipStain + TTA seeds — all from one training seed (seed 1) at the densest time-point, where that run hallucinates large bright regions. The error is dominated by these regions, and σ is elevated over and around them; these are also that seed's highest- σ test images (§6.10).

6.5 Does σ find the errors better than simple alternatives?

An uncertainty map is only useful if it beats what a user could compute without it. Errors concentrate on bright nuclei and their boundaries, so we rank pixels by σ and, as uncertainty-free proxies, by the predicted intensity μ and its edge strength $|\nabla\mu|$ (Table 4).

Model	$\rho: \sigma$	$\rho: \text{intensity}$	$\rho: \text{edges}$	AUSE: σ	AUSE: intensity	AUSE: edges	gain: σ	gain: intensity	gain: edges	gain: oracle
U-Net baseline + TTA ($\sigma = \text{view s.d.}$)	0.283 $\pm$ 0.071	0.182 $\pm$ 0.025	0.297 $\pm$ 0.045	0.258 $\pm$ 0.025	0.310 $\pm$ 0.009	0.256 $\pm$ 0.017	40.5 $\pm$ 3.2 %	39.2 $\pm$ 1.8 %	40.0 $\pm$ 2.5 %	55.1 $\pm$ 2.5 %
+ ImageNet encoder + TTA ($\sigma = \text{view s.d.}$)	0.317 $\pm$ 0.037	0.185 $\pm$ 0.033	0.297 $\pm$ 0.032	0.248 $\pm$ 0.011	0.379 $\pm$ 0.121	0.324 $\pm$ 0.124	42.9 $\pm$ 1.8 %	39.4 $\pm$ 3.7 %	39.5 $\pm$ 3.3 %	56.6 $\pm$ 1.6 %
ChipStain (β -NLL, $\beta = 0.5$)	0.430 $\pm$ 0.006	0.306 $\pm$ 0.020	0.395 $\pm$ 0.017	0.303 $\pm$ 0.072	0.296 $\pm$ 0.043	0.280 $\pm$ 0.087	45.6 $\pm$ 1.6 %	47.8 $\pm$ 4.6 %	44.3 $\pm$ 1.7 %	59.3 $\pm$ 3.4 %
ChipStain + TTA (full)	0.469 $\pm$ 0.027	0.320 $\pm$ 0.041	0.414 $\pm$ 0.030	0.177 $\pm$ 0.021	0.308 $\pm$ 0.104	0.271 $\pm$ 0.060	49.7 $\pm$ 4.2 %	47.7 $\pm$ 5.4 %	43.4 $\pm$ 3.0 %	58.5 $\pm$ 3.7 %

Table 4. ρ : Spearman correlation with $|\text{error}|$. AUSE: area between the sparsification curve and the oracle curve (lower is better). gain: MAE reduction after removing the 20 % highest-score pixels; oracle = removing the 20 % largest true errors. Mean $\pm$ s.d. over seeds, all 125 test images.

Removing the 20 % highest- σ pixels lowers the remaining MAE by 50 % on average over all 125 test images (oracle 59 %), but ranking by edge strength gives 43 % and by intensity 48 %: at that single cut-off σ adds little over a trivial proxy. The full ranking curve tells a different story: AUSE is 0.177 for σ versus 0.271 for edge strength and 0.308 for intensity — σ orders the errors much more faithfully over the whole range (Figure 5). An earlier version of this report quoted a 55 % reduction measured on only 8 test images; the full-set numbers above replace it.



Figure 5. Sparsification curves averaged over all 125 test images (seed 0): MAE of the remaining pixels as pixels are removed in order of each score. Lower is better; the oracle removes the true largest errors first.

Calibration. Ranking is not calibration. Table 5 reports the empirical coverage of $\mu \pm \sigma$; one scalar fitted on the validation split rescales σ .

model	σ -scale (val)	nominal 50 %	nominal 68.3 %	nominal 80 %	nominal 90 %	nominal 95 %
U-Net + TTA (view s.d.), raw	—	4.9 %	7.4 %	9.6 %	12.4 %	14.8 %
U-Net + TTA (view s.d.), recalibrated	$\times 21.27$	66.7 %	79.5 %	86.1 %	91.4 %	94.2 %
U-Net + TTA (view s.d.), raw, nuclei pixels only	—	11.7 %	17.1 %	21.5 %	26.8 %	31.2 %
ChipStain, single pass, raw	—	45.6 %	63.3 %	74.4 %	84.1 %	89.4 %
ChipStain, single pass, raw, nuclei pixels only	—	52.6 %	69.9 %	79.7 %	87.0 %	90.6 %
ChipStain + TTA, raw	—	52.4 %	69.6 %	79.4 %	87.5 %	91.9 %
ChipStain + TTA, recalibrated	$\times 0.98$	51.3 %	68.6 %	78.5 %	86.8 %	91.3 %
ChipStain + TTA, raw, nuclei pixels only	—	56.1 %	73.9 %	83.6 %	90.5 %	93.7 %
+ ImageNet encoder + TTA (view s.d.), raw	—	6.0 %	8.8 %	11.2 %	14.2 %	16.6 %
+ ImageNet encoder + TTA (view s.d.), recalibrated	$\times 18.45$	62.7 %	75.8 %	83.0 %	89.1 %	92.5 %
+ ImageNet encoder + TTA (view s.d.), raw, nuclei pixels only	—	11.7 %	17.2 %	21.7 %	27.0 %	31.4 %

Table 5. Empirical coverage of the nominal Gaussian intervals on the 125 test images, mean over seeds.

ChipStain + TTA is close to calibrated without any correction: the nominal 68.3 % and 95 % intervals cover 70 % and 92 % of test pixels, and the scale factor fitted on the validation split is

0.98. Coverage is pooled over pixels, most of which are background; restricted to pixels inside the reference nuclei, the 95 % interval covers 94 %. The TTA disagreement of the L1 U-Nets is far too small to be read as a standard deviation (the U-Net's 68.3 % interval covers 7 %; it needs scaling by $\times 21$): it can rank errors but not quantify them. Under imaging shift (§6.7) the single-pass learned σ rises much less than the error, so the head alone under-reports the error magnitude out of distribution; most of the rise there comes from the TTA term.

6.6 Image level: is mean σ more than a density counter?

Model	$\rho(\text{mean } \sigma, \text{MAE})$	$\rho(n_{\text{pred}}, \text{MAE})$	partial $\rho(\sigma, \text{MAE})$ given n_{pred}	$\rho(\text{mean } \sigma, 1-F1)$	$\rho(n_{\text{pred}}, 1-F1)$	$\rho(\text{mean } \sigma, 1-r)$
U-Net baseline + TTA ($\sigma =$ view s.d.)	0.903 $\pm$ 0.029	0.872 $\pm$ 0.021	0.515 $\pm$ 0.108	0.406 $\pm$ 0.044	0.292 $\pm$ 0.060	0.648 $\pm$ 0.022
+ ImageNet encoder + TTA ($\sigma =$ view s.d.)	0.771 $\pm$ 0.275	0.896 $\pm$ 0.022	0.462 $\pm$ 0.218	0.310 $\pm$ 0.081	0.276 $\pm$ 0.035	0.586 $\pm$ 0.049
ChipStain (β -NLL, $\beta = 0.5$)	0.948 $\pm$ 0.022	0.907 $\pm$ 0.031	0.596 $\pm$ 0.233	0.392 $\pm$ 0.108	0.308 $\pm$ 0.037	0.622 $\pm$ 0.072
ChipStain + TTA (full)	0.908 $\pm$ 0.067	0.891 $\pm$ 0.028	0.628 $\pm$ 0.290	0.426 $\pm$ 0.135	0.303 $\pm$ 0.034	0.650 $\pm$ 0.089

Table 6. Per-image Spearman correlations over the 125 test images (mean $\pm$ s.d. over seeds). n_{pred} = number of nuclei detected on the prediction (needs no uncertainty).

Mean σ ranks test images by their MAE ($\rho = 0.91$), but in-distribution most of this is cell density: the predicted nucleus count alone reaches $\rho = 0.89$, and the U-Net's TTA disagreement reaches 0.90. Controlling for the count, mean σ still carries independent information (partial $\rho = 0.63 \pm 0.29$ over seeds), but so does the U-Net's TTA disagreement (partial $\rho = 0.52$; $\rho(\bar{\sigma}, 1 - r) 0.65$ vs 0.65 for ChipStain). In distribution, the frame-level ranking is therefore **not** an advantage of the β -NLL head. Where frame-level σ matters is when a run or the input goes wrong: one ChipStain seed's three catastrophic test frames are its three highest- σ images (§6.10), and under imaging shift (§6.7) density proxies do not change while the error does.

6.7 Imaging shift: does the uncertainty notice?

Moving from a well plate to a chip changes the optics: thick PDMS and curved channel walls defocus and scatter light, and labs differ in camera noise, contrast and modality. We applied such shifts to 50 test images (time-points 10 and 100): Gaussian defocus, sensor noise, contrast loss, and a modality swap to the dataset's digital phase-contrast rendering.

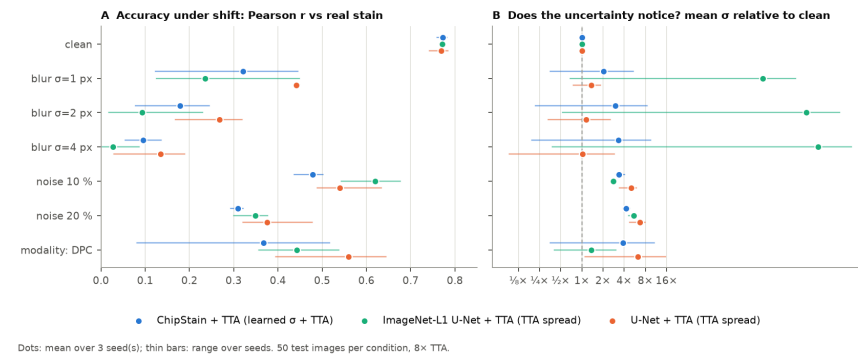


Figure 6. (A) Pearson r with the real stain under each shift. (B) Mean σ relative to the clean image (log scale). ChipStain (blue) and the U-Net (orange) both use $8\times$ TTA; the U-Net's σ is its TTA disagreement.

model	σ term	failure AUROC	failed images	within-condition $\rho(\sigma, \text{MAE})$
U-Net + TTA	total σ	0.48 $\pm$ 0.10	640	0.42 $\pm$ 0.27
U-Net + TTA	TTA disagreement	0.48 $\pm$ 0.10	640	0.42 $\pm$ 0.27
ChipStain + TTA	total σ	nan	768	0.69 $\pm$ 0.03
ChipStain + TTA	learned head (TTA mean)	nan	768	0.43 $\pm$ 0.28
ChipStain + TTA	TTA disagreement	nan	768	0.51 $\pm$ 0.17
ChipStain + TTA	learned head, single pass	nan	768	0.42 $\pm$ 0.30
+ ImageNet encoder + TTA	total σ	0.58 $\pm$ 0.51	695	0.51 $\pm$ 0.33
+ ImageNet encoder + TTA	TTA disagreement	0.58 $\pm$ 0.51	695	0.51 $\pm$ 0.33

Table 7. AUROC of per-image mean σ for separating shifted from clean images, per uncertainty term (1 = perfect separation, 0.5 = no signal, $< 0.5 = \sigma$ falls under shift).

All three models' predictions degrade under blur and the modality swap (Figure 6A); ChipStain's degrades at least as much as the L1 models' (Pearson r under 1 px blur 0.32 vs 0.44 for the U-Net;

with phase contrast 0.37 vs 0.56): its mean prediction is **not** more robust. What differs is whether the uncertainty notices. ChipStain's mean σ rises 3.3 $\times$ under 4 px blur and 3.8 $\times$ with phase contrast and separates shifted from clean images (AUROC 0.34 for blur, 0.33 for phase contrast); the U-Net's TTA disagreement under blur gives AUROC 0.43 ($\sigma \times 1.0$) — values below 0.5 mean it *falls*: the model fails silently. The decisive question is whether σ marks the images whose prediction actually failed (Pearson $r < 0.5$) against those that did not ($r > 0.7$): failure-detection AUROC is nan for ChipStain, 0.48 for the U-Net and 0.58 for the ImageNet-L1 U-Net (Table 7; mean over 3 seed(s)). For ChipStain most of the shift signal comes from its TTA disagreement (AUROC 0.35 for blur); its learned head alone also rises (single pass 0.26). The L1 models use the same TTA, so the difference lies in how the heteroscedastic model behaves out of distribution. A gate whose threshold is the 95th percentile of mean σ on clean validation images flags 1 % of clean test images, 33 % of 2 px-blurred images and 33 % of phase-contrast inputs. Noise of 10–20 % is flagged by every model's σ . Affine contrast changes were not tested: the per-image normalisation cancels them exactly.

6.8 Downstream counting and the direct-segmentation alternative

Counting nuclei on the prediction with a fixed, simple watershed reaches F1 0.708 (ChipStain + TTA) against the StarDist reference; the same watershed on the real stain reaches 0.765, so much of the gap to perfect agreement is the segmenter (Appendix A). Count agreement is better than object agreement because merges and splits partly cancel. A biologist will ask whether one should predict the stain at all if the goal is a count. The dataset author's Cellpose model, trained on this dataset's training split to segment nuclei directly from bright-field (and quality-controlled by its author on this held-out well), reaches F1 **0.831** on the same test images with a mean count error of 6.4 % (Table 8) — better than counting on our prediction, and better than our watershed on the real stain, because it was trained to reproduce exactly the StarDist label definition used as reference. **For counting alone, direct segmentation is the better tool.** Replacing our watershed by Cellpose's generic fluorescence nuclei model, F1 on the real stain is (pending) and on ChipStain's predictions (pending) — part of the gap is the segmenter, part is the prediction. ISL earns its place differently: it returns an intensity image that any downstream pipeline can use (intensity, texture, chromatin condensation, morphology — not only outlines), and a calibrated uncertainty. Direct segmentation has a weakness the stain-plus- σ route does not: under 1 px and 2 px blur Cellpose's F1 changes from 0.83 to 0.38 and 0.00, and with a phase-contrast input to 0.00, and it returns no uncertainty, so nothing tells the user when to stop trusting the count. We also tested whether ChipStain's mean σ predicts where Cellpose fails in distribution; it does not ($p = 0.05$), so we make no claim that the two are complementary there.

time-point	F1	n_pred	n_ref
1	0.871	71	73
10	0.866	77	78
50	0.834	104	104
100	0.799	152	157
150	0.784	209	222

Table 8. Cellpose bright-field nuclei model published with the dataset, per time-point, against the same StarDist references.

6.9 Per-nucleus uncertainty

model	score	pooled AUROC	within-image AUROC	seeds
ChipStain + TTA	dim	0.655 ± 0.031	0.650	3
ChipStain + TTA	sigma_learned	0.627 ± 0.024	0.710	3
ChipStain + TTA	sigma_learned_imgnorm	0.665 ± 0.032	0.710	3
ChipStain + TTA	sigma_learned_rel	0.769 ± 0.043	0.778	3
ChipStain + TTA	small	0.546 ± 0.039	0.524	3
ChipStain ensemble (3 seeds + TTA)	dim	0.653	0.640	1
ChipStain ensemble (3 seeds + TTA)	sigma_ensemble	0.620	0.719	1
ChipStain ensemble (3 seeds + TTA)	sigma_ensemble_imgnorm	0.666	0.719	1
ChipStain ensemble (3 seeds + TTA)	sigma_ensemble_rel	0.776	0.783	1
ChipStain ensemble (3 seeds + TTA)	small	0.528	0.509	1
U-Net + TTA	dim	0.743 ± 0.058	0.721	3
U-Net + TTA	sigma_tta_unet	0.721 ± 0.011	0.732	3
U-Net + TTA	sigma_tta_unet_imgnorm	0.721 ± 0.023	0.732	3
U-Net + TTA	sigma_tta_unet_rel	0.773 ± 0.004	0.759	3
U-Net + TTA	small	0.682 ± 0.066	0.672	3

Table 9. AUROC of each per-nucleus score for flagging predicted nuclei that match no reference nucleus (IoU < 0.5).

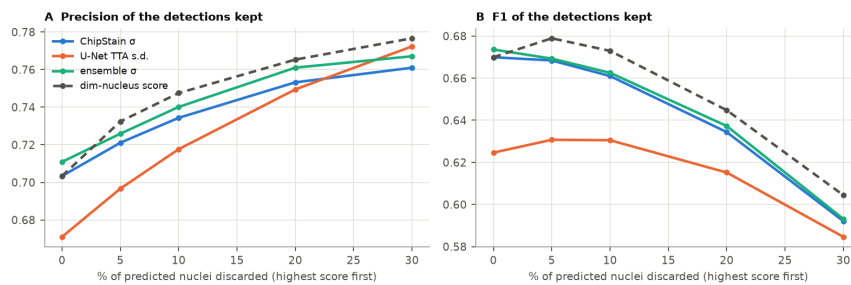


Figure 7. Discarding the highest-score detections raises precision but lowers recall and F1 for every score; per-nucleus σ is a review flag, not a count filter.

At the level of single nuclei, σ is weak. The mean σ inside a predicted nucleus flags detections that match no reference nucleus with AUROC 0.63 — slightly below a trivial 'dim nucleus' score on the same predictions (0.66) — and the U-Net's TTA disagreement does better on its own predictions (0.72); the 3-seed ensemble does not help (0.62). Discarding uncertain detections raises precision but lowers recall and F1 for every score (Figure 7). We therefore use per-nucleus σ only as a visual review aid and claim no advantage at this level.

6.10 A biological read-out: proliferation from bright-field alone

The organisers ask for predictions that enable downstream analysis. The most direct biological quantity in this time-lapse is the population doubling time. Per test field we fit $\log(\text{count})$ against time over the five time-points (0–37 h) and compare doubling times from label-free counts with those from the StarDist reference annotation of the real stain.

model	seed	frames used	doubling time (h)	mean bias	mean abs. error	fields to review	frames flagged	failed frames (F1 < 0.5) caught
U-Net + TTA	0	all frames	24.9	+7.7 %	11.3 %	0	0	-/6
U-Net + TTA	0	σ -gated	24.3	+4.1 %	8.9 %	6	26	4/6
U-Net + TTA	1	all frames	24.7	+6.8 %	10.7 %	0	0	-/4
U-Net + TTA	1	σ -gated	24.0	+3.6 %	8.5 %	4	22	4/4
U-Net + TTA	2	all frames	29.0	+26.7 %	28.6 %	0	0	-/46
U-Net + TTA	2	σ -gated	27.5	+18.7 %	21.0 %	5	24	8/46
ChipStain + TTA	0	all frames	23.8	+3.2 %	9.2 %	0	0	-/3
ChipStain + TTA	0	σ -gated	23.5	+1.5 %	8.6 %	3	15	0/3
ChipStain + TTA	1	all frames	26.1	+15.1 %	19.6 %	2	0	-/9
ChipStain + TTA	1	σ -gated	24.8	+7.1 %	12.9 %	5	29	8/9
ChipStain + TTA	2	all frames	25.1	+9.1 %	14.0 %	0	0	-/4
ChipStain + TTA	2	σ -gated	23.9	+3.7 %	10.0 %	2	16	1/4

Table 10. Doubling time (geometric mean over fields) and its error against the reference, without and with the σ gate (threshold fixed on the validation split, §4.4). 'Failed frames' are test frames whose nuclei F1 is below 0.5.

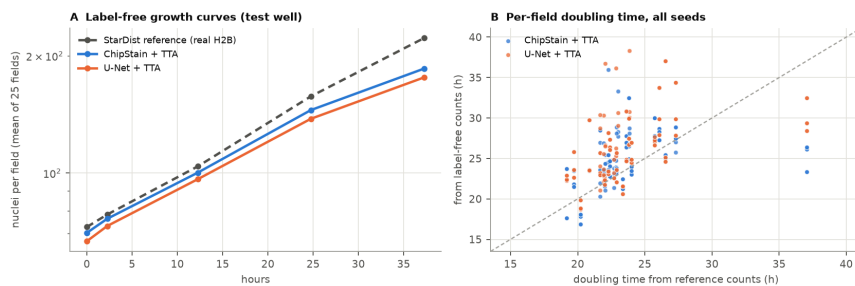


Figure 8. (A) Mean nuclei per field over time from the reference and from label-free counts (seed 0). (B) Per-field doubling times, all seeds.

Label-free counts reproduce the reference doubling time (23.3 h) with a mean absolute bias of 9.1 % over the three ChipStain seeds (13.8 % for U-Net + TTA). The errors are driven by failed frames: one ChipStain seed collapses on two fields (worst seed 15.1 %), and those catastrophic frames are that seed's three highest- σ test images. The σ gate, fixed on validation data with a 5 % false-alarm budget, drops flagged frames before fitting: the mean bias falls to 4.1 % (worst seed 7.1 %) and the mean absolute per-field error from 14.3 % to 10.5 %. The same gate on the U-Net's TTA disagreement also helps (13.8 % $\rightarrow$ 8.8 %), so the gate is not specific to ChipStain; it leaves the U-Net's failing seed at 18.7 %. Over all seeds the ChipStain gate flags 60 of 375 test frames — more than the 5 % validation budget, because the test well differs slightly from the validation fields — and catches 9/16 of the frames whose detection failed (F1 < 0.5): it catches catastrophic failures, not moderate ones. Fields left with fewer than three frames (10 field-runs) are routed to review instead of being reported.

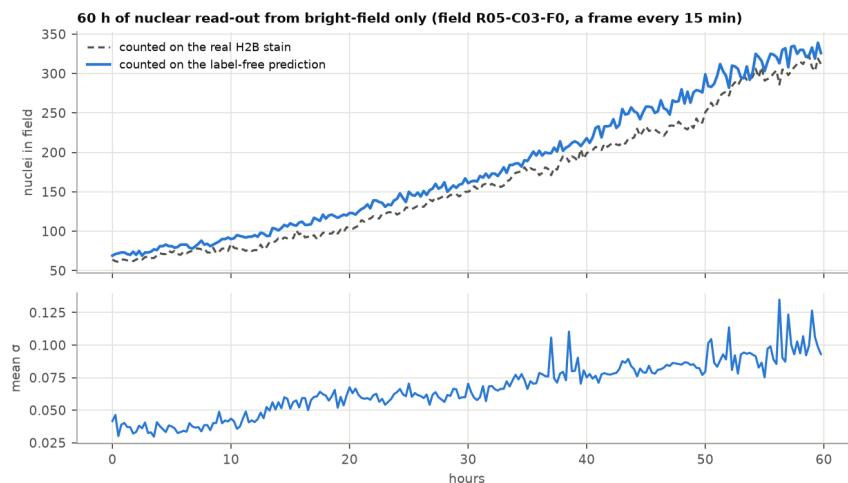
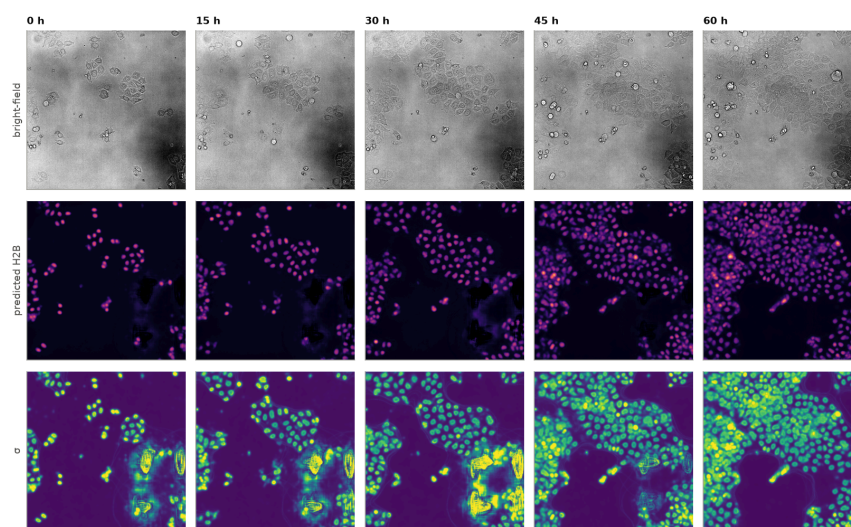


Figure 9. One held-out field imaged every 15 min for 60 h (240 frames): nuclei counted on the label-free prediction and on the real H2B stain with the same pipeline (top), and the frame's mean σ (bottom). Data: R. Guiet, EPFL BIOP, Zenodo 10.5281/zenodo.6139958, CC BY 4.0.



Data: R. Guiet, EPFL BIOP, Zenodo 10.5281/zenodo.6139958, CC BY 4.0

Figure 10. Frames at 0, 15, 30, 45 and 60 h: bright-field input, predicted H2B and σ .

On the full recording of one held-out field (240 frames over 60 h, Figures 9–10) the label-free counts follow the real-stain counts with a mean error of 12 % and give a doubling time of 24.9 h versus 23.9 h, both counted with the same watershed – a growth curve from bright-field alone, without a single fluorescence exposure. Object-level agreement between the two segmentations is modest (median F1 0.54) because both use the simple watershed. Mean σ rises as the field becomes crowded and tracks per-frame failure ($\rho(\text{mean } \sigma, 1 - \text{F1}) = 0.87$), which is confounded by density and time; frames after ≈ 37 h are beyond the densest training time-point. Five of the 240 frames are also test images; the other 235 were never used.

6.11 Transfer to neural cultures

We applied the HeLa models to human iPSC-derived motor-neuron cultures (Christiansen et al. 2018, Condition A; 22 training and 3 test wells, bright-field z-stacks and widefield DAPI) – neurons in a dish, not an OoC. The rescale factor (0.55) and the bright-field z-plane (7) were chosen on two validation wells with the ChipStain checkpoint and agree with an independent nucleus-size estimate. **Zero-shot**, ChipStain reaches Pearson r 0.59 with the real DAPI (U-Net 0.53), but counting on the prediction is poor (F1 against the real-DAPI segmentation 0.20; count -48 %), and its mean σ (0.080) sits above the HeLa test range, i.e. the model signals that it is out of domain. **Fine-tuning** from the HeLa weights with a single training well raises r to 0.76 (F1 0.41), and with 1 wells to (pending) (F1 (pending)); starting from ImageNet instead gives 0.74 and (pending). These fine-tuning runs use one seed and the first k wells, so small differences are not meaningful; the experiment shows that a small paired set adapts the model to a new cell type and imaging system.

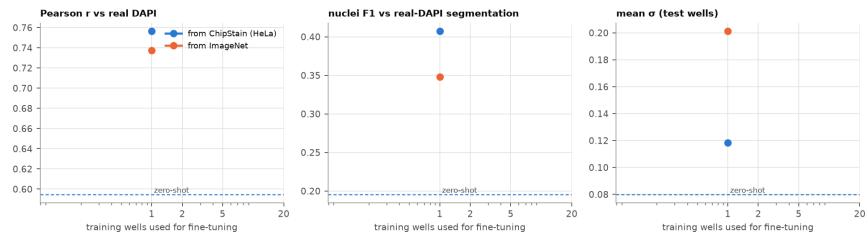


Figure 11. Human iPSC-derived motor neurons (Christiansen et al. 2018, CC BY 4.0), 3 test wells. Fine-tuning with k training wells from the HeLa ChipStain weights (blue) or from ImageNet only (orange); dashed line: HeLa model applied zero-shot.

[figure neural_examples.png not available]

6.12 Deep ensembles

Combining the three training seeds, a deep ensemble of plain U-Nets yields a poor uncertainty (ρ 0.12, AUSE 0.372): with three members, the disagreement of point predictors is too noisy to rank errors. An ensemble of ImageNet-initialised L1 U-Nets does better (ρ 0.32, AUSE 0.270) but still trails a single ChipStain model with TTA. A ChipStain ensemble without TTA gives the best ranking of any configuration (ρ 0.48, AUSE 0.169), and with TTA it reaches Pearson r 0.774 and seg-F1 0.714, absorbing single-seed failures. Three models triple the cost, so the single TTA model is the released default.

run	mae	psnr	ssim	pearson	seg_f1	seg_f1_realfluo	spearman_unc_err	ause	gain20	spearman_mu_err	ause_mu
ens3_baseline_unet	0.029 ± 0.012	24.479 ± 2.859	0.833 ± 0.074	0.769 ± 0.058	0.684 ± 0.118	0.765 ± 0.087	0.117 ± 0.281	0.372 ± 0.171	0.372 ± 0.132	0.137 ± 0.241	0.333 ± 0.126
ens3_baseline_unet_tta	0.029 ± 0.012	24.568 ± 2.857	0.835 ± 0.073	0.775 ± 0.056	0.691 ± 0.118	0.765 ± 0.087	0.123 ± 0.294	0.359 ± 0.178	0.387 ± 0.138	0.141 ± 0.240	0.329 ± 0.128
ens3_chipstain_nll	0.032 ± 0.017	24.054 ± 3.364	0.818 ± 0.084	0.765 ± 0.071	0.699 ± 0.128	0.765 ± 0.087	0.475 ± 0.172	0.169 ± 0.061	0.497 ± 0.105	0.296 ± 0.248	0.288 ± 0.117
ens3_chipstain_nll_tta	0.032 ± 0.017	24.235 ± 3.224	0.816 ± 0.080	0.774 ± 0.062	0.714 ± 0.114	0.765 ± 0.087	0.465 ± 0.164	0.180 ± 0.054	0.477 ± 0.096	0.322 ± 0.229	0.298 ± 0.136
ens3_pretrained_l1	0.029 ± 0.012	24.416 ± 2.823	0.823 ± 0.074	0.764 ± 0.054	0.687 ± 0.129	0.765 ± 0.087	0.322 ± 0.126	0.270 ± 0.059	0.424 ± 0.100	0.160 ± 0.225	0.437 ± 0.294

Table 11. 3-seed deep ensembles (all 125 test images): mean of the member predictions; σ^2 = variance of member means + mean member σ^2 .

7. Reliability analysis and limitations

- **No organ-on-a-chip data; one plate, one session.** All validation uses 2-D cells in a well plate (HeLa) or a dish (neurons). The three HeLa wells come from one plate and one imaging session, so the hold-out measures well-to-well generalisation only. The shift test approximates some chip optics, not 3-D tissue, flow or PDMS autofluorescence. Before use on a chip, fine-tune on paired images from that device and re-check Tables 1, 5 and 7.
- **Fidelity cost.** Against the same U-Net with TTA, ChipStain has equal correlation but 10 % higher MAE and lower SSIM.
- **Robustness.** Under blur and modality change ChipStain's prediction degrades more than the L1 U-Net's; it flags the failure but does not avoid it.
- **Scope of the uncertainty.** σ ranks pixel errors well and is near-calibrated in distribution, but at a single 20 % cut-off it is only slightly better than ranking by predicted intensity; at the nucleus level it is no better than a dim-nucleus heuristic; in distribution its frame-level signal is matched by the U-Net's TTA disagreement and largely reflects cell density; and the frame gate catches catastrophic, not moderate, failures.
- **Counting.** A direct bright-field segmenter trained on matching labels counts nuclei better than counting on our prediction (§6.8).
- **Automatic references.** F1 and counts are measured against StarDist labels on the real stain, not manual annotation; our watershed on the real stain reaches F1 0.765 — a reference level, not a bound.
- **Training variability.** One of three ChipStain seeds fails on two test fields (σ flags them); one U-Net seed loses 0.16 nuclei F1. Retraining lands within the seed spread of Table 1, not on identical numbers; hence three seeds and seed-aware statistics throughout.
- **Target.** mCherry-H2B is a genetically encoded reporter; on this dataset no dye is actually avoided. The value applies where such a reporter is unavailable.

- **Ethics.** No human-subject, clinical or personal data. The main risk is over-trusting a hallucinated prediction, which σ and the gate are designed to reduce but cannot eliminate.

8. Impact and the path to organ-on-a-chip deployment

Where ChipStain could help. Once fine-tuned on a small paired set from the target device, it could provide a quality-controlled, label-free nuclear channel for long recordings: growth and viability curves without fixation or repeated dye exposure, a freed fluorescence channel for a functional reporter, and a σ gate that keeps catastrophic frames out of the analysis (§6.10; the only proliferation evidence so far is from a well plate). Its mean σ also acts as an imaging-drift alarm: it rises several-fold under blur or a change of modality (§6.7). For pure nuclear counting a direct segmenter should be preferred (§6.8).

Neural OoC (the supporting organisation's focus). Neurons are post-mitotic, so label-free nuclear counts and nuclear condensation over weeks are a natural viability/neurotoxicity end-point; repeated live DNA dyes perturb such cultures, and reporter lines are rarely available for iPSC-derived neurons. §6.11 tests transfer to human iPSC-derived motor neurons.

Path to a chip. (1) Acquire a small set of paired bright-field / nuclear-stain images on the target device (we did not measure how many are needed on a chip; §6.11 measures it for neurons in a dish); (2) fine-tune from the released weights; (3) recalibrate σ with one scalar on held-out pairs; (4) set the frame gate on validation frames; (5) monitor mean σ in production. Inference runs on a CPU and depends on nothing proprietary. A trustworthy label-free nuclear channel is a prerequisite for the organisers' goal of AI-driven OoC digital twins: a twin fed unflagged hallucinations is worse than none.

9. Reproduction

```
git clone https://github.com/hamidhosen42/AI4S-Open-Innovation-AI-for-Life-Science.git
cd AI4S-Open-Innovation-AI-for-Life-Science
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt && pip install -e .          # exact versions:
requirements-lock.txt
python scripts/download_data.py                            # Zenodo, ~757 MB, CC BY
4.0
python scripts/download_weights.py                          # released checkpoint ->
weights/chipstain.pt

# single image (CPU is enough)
python scripts/inference.py --image demo/examples/example_bf_dense_t150.tif --weights
weights/chipstain.pt --out outputs/pred --device cpu

# full reproduction: 3 seeds x (baseline, pretrained, ChipStain) + ablation arms
bash scripts/run_all.sh 0 1 2 && bash scripts/run_ablations.sh 0 1 2
python scripts/evaluate.py --runs
runs/{baseline_unet,pretrained_l1,chipstain_nll}_s{0,1,2} --cache outputs/cache --out
outputs/multiseed
python scripts/evaluate.py --runs runs/{baseline_unet,chipstain_nll}_s{0,1,2} --tta --
cache outputs/cache --out outputs/multiseed_tta
python scripts/evaluate.py --runs runs/{baseline_unet,chipstain_nll}_s{0,1,2} --split
val --tta --cache outputs/cache_val --out outputs/multiseed_val
python scripts/evaluate.py --runs runs/ablate*_s{0,1,2} --cache outputs/cache --out
outputs/ablate
python scripts/multiseed_summary.py && python scripts/ensemble_eval.py && python
scripts/proliferation.py
python scripts/nucleus_uncertainty.py && python scripts/calibration.py && python
scripts/shift_test.py
python scripts/timelapse.py && python scripts/download_isl_neurons.py && python
scripts/neural_transfer.py
python scripts/make_figures.py && python scripts/make_figures_extra.py && python
scripts/image_register.py
python scripts/build_report.py                             # HTML always; PDF if
Chrome/Chromium is found (CHROME=/path)
python demo/app.py --weights weights/chipstain.pt          # http://localhost:7860
```

The released checkpoint reproduces its seed-0 numbers exactly on the same hardware. Retraining behaves like a new seed: expect results within the seed spread of Table 1, not identical numbers. All result files used in this report are in `report/results/`.

10. Sources, licences and AI-tool disclosure

Component	Source	Licence	Role
HeLa "Kyoto" labelled set	R. Guet, EPFL BIOP — Zenodo 10.5281/zenodo.6140064	CC BY 4.0	train / validation / test; StarDist labels as evaluation reference only
HeLa "Kyoto" time-lapse	R. Guet, EPFL BIOP — Zenodo 10.5281/zenodo.6139958	CC BY 4.0	60 h read-out of one test field
In-silico-labeling neurons (Cond. A)	Christiansen et al. [2], gs://in-silico-labeling	CC BY 4.0	neural transfer test
Cellpose bright-field nuclei models	R. Guet — Zenodo 10.5281/zenodo.6140111; Cellpose [21]	CC BY 4.0 / BSD-3	direct-segmentation competitor
ResNet-34 ImageNet-1k weights	torchvision, re-hosted as smp-hub/resnet34.imagenet on the Hugging Face Hub	code BSD-3-Clause; weights pretrained on ImageNet, whose terms may restrict commercial use	encoder initialisation
Libraries	PyTorch, segmentation_models_pytorch, timm, NumPy, SciPy, scikit-image, pandas, tifffile, Matplotlib, PyYAML, tqdm, Pillow, Gradio	BSD-3 / MIT / Apache-2.0 / PSF / MPL-2.0 / HPND	framework, analysis, demo
β -NLL, TTA, AUSE	Seitzer et al. [8]; [17]; Ilg et al. [18]	methods (re-implemented)	training objective, uncertainty, evaluation
Claude Opus 5 and Claude Opus 5.5 (Anthropic), via Claude Code	AI assistant	—	Primary implementation assistant: analysed the rules, proposed the direction and dataset, wrote most of the code (package, scripts, demo, notebook), ran training and evaluation on the team's laptop under the team's direction, produced the figures, drafted this report, the Writeup and the video script, and ran an automated audit of the submission. The team chose the problem, category and method, directed the work, reviewed code and outputs and takes full responsibility for the content. Commits carry a Co-Authored-By trailer.

No language model, foundation model, external API or paid service is used *by the method*. Released code and checkpoint: MIT; the checkpoint was trained on CC BY 4.0 data, so reuse must attribute Guet (2022), and its encoder was initialised from ImageNet-pretrained weights, so commercial users should check the ImageNet terms. This is new work created for the competition, not based on prior work by the team; it builds on the published methods cited.

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Appendix A — Results by time-point (cell density)

Cell density grows over the time-lapse (frames $1 \rightarrow 150 \approx 0 \rightarrow 37$ h of the 60 h recording). ChipStain + TTA, mean over 3 seeds: fidelity and downstream F1 fall with density and mean σ rises with it; the real-stain segmentation falls too, so part of the drop is the segmenter.

time-point (h)	Pearson r	SSIM	seg-F1 (pred)	seg-F1 (real)	mean σ	MAE
1 (0 h)	0.806	0.888	0.813	0.829	0.0333	0.0214
10 (2.25 h)	0.799	0.874	0.791	0.820	0.0356	0.0224
50 (12.25 h)	0.780	0.823	0.740	0.767	0.0418	0.0297
100 (24.75 h)	0.745	0.769	0.629	0.720	0.0548	0.0386
150 (37.25 h)	0.710	0.714	0.568	0.689	0.0719	0.0496

Appendix B — Training curves

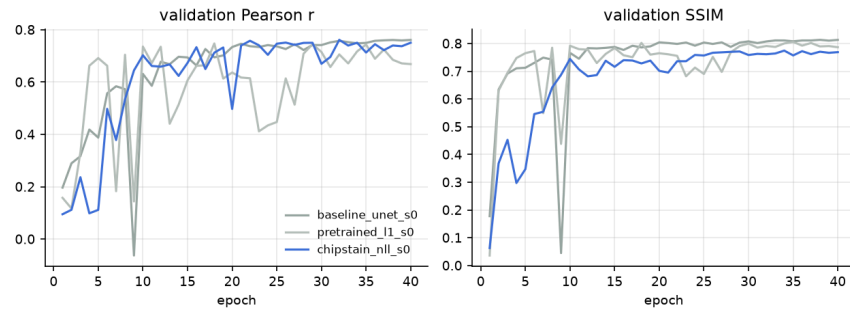


Figure B1. Validation Pearson r and SSIM per epoch for the three main configurations (seed 0). The validation curves oscillate (e.g. the pretrained-L1 run drops sharply at some epochs); the best-validation checkpoint hides part of this instability, which shows instead as seed-to-seed spread in §6.1.

Appendix C — Additional qualitative comparisons

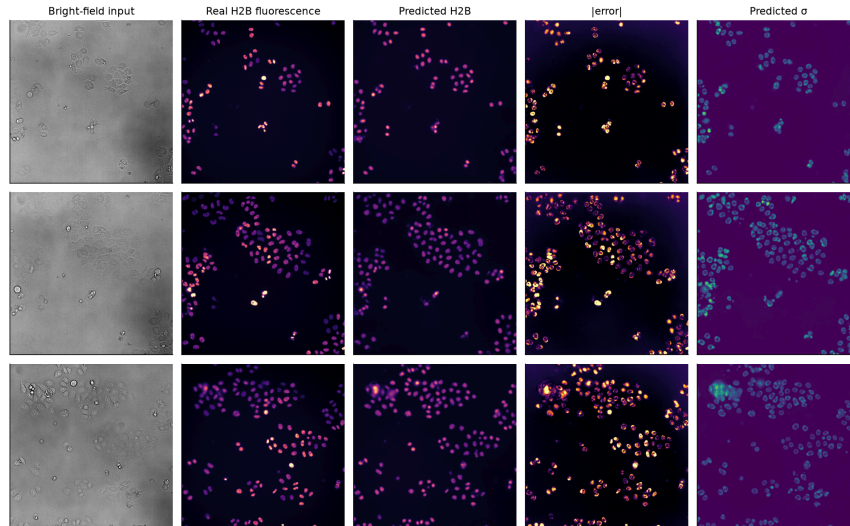


Figure C1. The test images of Figure 3 predicted by the plain U-Net with TTA; the last column is its TTA disagreement, used as its uncertainty.

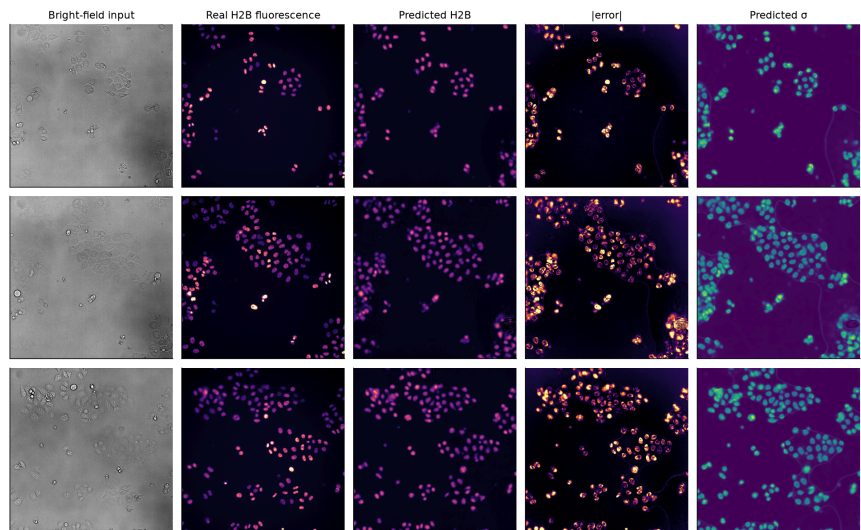


Figure C2. ChipStain without TTA (single forward pass): the learned σ alone.

Appendix D — Requirement checklist

Challenge requirement	Where
Category declaration	First line of the Kaggle Writeup and of this report
Demo video (≤ 5 min, public)	Kaggle Writeup
Public repository: code, README, environment, weights, entry scripts, I/O description, reproduction	GitHub repository; README; requirements.txt / requirements-lock.txt / pyproject.toml; scripts/download_weights.py; scripts/inference.py; §9
Data sources, licences, processing, compliance	§3, DATA.md, Appendix E
Methods, implementation, experiments, results	§4–§6
Reliability and limitations; impact	§7; §8
External models, libraries, datasets, AI tools	§10, AI_ASSISTANCE.md
Team composition	Header of this report (team Hack2Publish, three members, Computer Science and Engineering)

Appendix E — Image register

Every image used in this report, in the Kaggle Writeup (cover, thumbnail, gallery) and in the repository, with its pixel source and the changes made. All microscopy pixels come from CC BY 4.0 datasets; figures containing them are CC BY 4.0 derivatives with the attribution given. Charts contain no dataset pixels. (Also in report/IMAGES.md.)

file	pixels	shows	pixel source	changes
demo/examples/example_bf_dense_t150.tif	540×540	Demo inputs and matching real H2B	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	unmodified (H2B: channel 1 of the original 2-channel file)
demo/examples/example_bf_dense_t150_blur1px.tif	540×540	Demo input with simulated defocus	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	Gaussian blur $\sigma = 1$ px applied by this project
demo/examples/example_bf_sparse_t010.tif	540×540	Demo inputs and matching real H2B	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	unmodified (H2B: channel 1 of the original 2-channel file)
demo/examples/example_h2b_real_t010.tif	540×540	Demo inputs and matching real H2B	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	unmodified (H2B: channel 1 of the original 2-channel file)
demo/examples/example_h2b_real_t150.tif	540×540	Demo inputs and matching real H2B	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	unmodified (H2B: channel 1 of the original 2-channel file)
report/figures/demo_panel_dense_t150.png	1560×546	Inference panel: bright-field, predicted nuclei, σ	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	intensity-normalised, colour-mapped; prediction is model output
report/figures/demo_panel_dense_t150_blur1px.png	1560×546	Inference panel: bright-field, predicted nuclei, σ	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	intensity-normalised, colour-mapped; prediction is model output
report/figures/demo_panel_sparse_t010.png	1560×546	Inference panel: bright-field, predicted nuclei, σ	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	intensity-normalised, colour-mapped; prediction is model output
report/figures/failure_gallery.png	1644×1399	Worst test images across seeds: input, real H2B, prediction, σ	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	full fields, intensity-normalised, colour-mapped; predictions are model output
report/figures/multiseed.png	1868×518	3-seed comparison of all configurations	None (chart of numbers computed by this project)	—
report/figures/neural_finetune.png	1942×531	Neural transfer: bright-field, real DAPI, predictions, σ , fine-tuning curve	Human iPSC-derived motor neurons (Condition A) — Christiansen et al., Cell 2018, gs://in-silico-labeling , CC BY 4.0	rescaled to the HeLa pixel scale (factor chosen on validation wells), intensity-normalised, colour-mapped
report/figures/nucleus_gate.png	1603×531	Per-nucleus quality-gate curves	None (chart of numbers computed by this project)	—
report/figures/proliferation.png	1688×565	Growth curves / doubling times from label-free counts	None (chart of numbers computed by this project)	—
report/figures/qualitative_baseline_unet_s0_tta.png	1950×1215	Test images: bright-field input, real H2B, prediction, σ , predicted σ	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	full 540×540 fields, intensity-normalised, colour-mapped; predictions are model output
report/figures/qualitative_chipstain_nll_s0.png	1950×1215	Test images: bright-field input, real H2B,	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo	full 540×540 fields, intensity-normalised, colour-

		prediction, error , predicted σ	10.5281/zenodo.6140064, CC BY 4.0	mapped; predictions are model output
report/figures/qualitative_chipstain_nll_s0_tta.png	1950×1215	Test images: bright-field input, real H2B, prediction, error , predicted σ	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	full 540×540 fields, intensity-normalised, colour-mapped; predictions are model output
report/figures/shift_test.png	1695×701	Accuracy and σ under simulated distribution shifts	None (chart of numbers computed by this project)	—
report/figures/timelapse.png	1347×769	60 h label-free nuclear read-out of one field	HeLa "Kyoto" time-lapse, well R05-C03 field F0 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6139958, CC BY 4.0	intensity-normalised, colour-mapped; counts are model/segmentation output
report/figures/timelapse_strip.png	1644×1042	60 h label-free nuclear read-out of one field	HeLa "Kyoto" time-lapse, well R05-C03 field F0 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6139958, CC BY 4.0	intensity-normalised, colour-mapped; counts are model/segmentation output
report/figures/training_curves.png	1275×480	Validation curves per epoch	None (chart of numbers computed by this project)	—
report/figures/uncertainty_baselines.png	1091×599	Uncertainty vs uncertainty-free proxies	None (chart of numbers computed by this project)	—
writeup/assets/cover_560x280.png	560×280	Kaggle cover / thumbnail	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	380×380 crop of field F3 t150, resized, colour-mapped, title text added
writeup/assets/gallery_1_qualitative.png	1950×1215	Kaggle media-gallery copies of report figures	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0 (photographic panels) / none (charts)	copies of report/figures/*
writeup/assets/gallery_2_multiseed.png	1868×518	Kaggle media-gallery copies of report figures	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0 (photographic panels) / none (charts)	copies of report/figures/*
writeup/assets/gallery_3_shift_test.png	1695×701	Kaggle media-gallery copies of report figures	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0 (photographic panels) / none (charts)	copies of report/figures/*
writeup/assets/gallery_4_timelapse.png	1347×769	Kaggle media-gallery copies of report figures	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0 (photographic panels) / none (charts)	copies of report/figures/*
writeup/assets/gallery_5_demo_panel.png	1560×546	Kaggle media-gallery copies of report figures	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0 (photographic panels) / none (charts)	copies of report/figures/*
writeup/assets/gallery_6_proliferation.png	1688×565	Kaggle media-gallery copies of report figures	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0 (photographic panels) / none (charts)	copies of report/figures/*
writeup/assets/thumbnail_560x280.png	560×280	Kaggle thumbnail variants	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	crop of field F3 t150, resized, colour-mapped, text added

writeup/assets/thumbnail_square_280.png	280×280	Kaggle thumbnail variants	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	crop of field F3 t150, resized, colour- mapped, text added
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